

National University of Computer and Emerging Sciences



Lab Manual 08

Programming Fundamentals

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Semester	Fall 2025

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Activities:

Note: Include input validation in programs wherever applicable. Don't use any build-in function.

1. Write a function that swaps the values of three variables in a circular manner (i.e., a gets b's value, b gets c's value, and c gets a's value). The user-defined function should not use 4th variable for swapping and the main function should display the values after swapping.

Example Input:

Enter three numbers: 10 20 30

Example Output:

Before swap: a = 10, b = 20, c = 30

After swap: a = 20, b = 30, c = 10

2. Create a program with two functions. One function checks whether a number entered by user is prime and returns a boolean result. Another function calculates the factorial of a prime number to store the result in parameters that has be printed in main() function. Remember, the program just need to print the factorial of those numbers that are prime.

Example Input:

Enter a number: 5

Example Output:

Factorial of 5 is 120.

3. Write a program that calculates the area and circumference of a circle. Define the following:
 - a. A global constant PI with the value 3.14159.
 - b. A function getRadius() that prompts the user to enter the radius of the circle and returns the radius.
 - c. A function calculateArea(float radius) that takes the radius as an argument, calculates the area using the formula $\text{area} = \text{PI} \times \text{radius}^2$, and returns the area.
 - d. To calculate radius^2 , you also have to define a function power(int base, int exponent) that returns the calculated result.
 - e. A function calculateCircumference(float radius) that calculates and returns the circumference using the formula $\text{circumference} = 2 \times \text{PI} \times \text{radius}$.
 - f. In the main function, call getRadius() to obtain the radius, then use calculateArea() and calculateCircumference() to compute and display the area and circumference.

Example Input:

Enter the radius of the circle: 5

Expected Output:

Area of the circle: 78.53975

Circumference of the circle: 31.4159

4. Write a program to calculate the monthly payment on a loan. Define the following:
 - a. Global variables principal, rate, and years to store the loan amount, interest rate, and loan term in years.
 - b. A function getLoanDetails() that prompts the user to enter the loan principal, interest rate, and number of years, storing these values in the global variables.
 - c. A function calculateMonthlyPayment() that calculates and returns the monthly payment using the formula:

$$\text{monthly payment} = \frac{\text{principal} * \left(\frac{\text{rate}}{12}\right)}{1 - \left(1 + \frac{\text{rate}}{12}\right)^{-\text{years} * 12}}$$

Note: Remember, you are not using any build-in function. If needed, you can define it yourself.

- d. In the main function, call getLoanDetails() to get loan details, then use calculateMonthlyPayment() to compute and display the monthly payment.

Example Input:

Enter loan amount: 10000

Enter annual interest rate (in decimal, e.g., 0.05 for 5%): 0.05

Enter loan term in years: 3

Expected Output:

Monthly payment: 299.71

5. Implement a program to calculate the income tax for individuals based on their income slabs. Create overloaded functions named CalculateTax to handle the following cases:
- For salaried individuals:** Pass their monthly salary and tax percentage as parameters. The default tax percentage is 15%.
 - For business owners:** Pass annual revenue and business expenses. Tax is calculated as a percentage of net profit (revenue - expenses).
 - For freelancers:** Pass their earnings and the number of dependents. Tax is reduced by 2% per dependent (max reduction 10%).

Function Prototypes:

- float CalculateTax(float monthlySalary, float taxPercentage = 15.0);
- float CalculateTax(float revenue, float expenses);
- float CalculateTax(float earnings, int dependents);

Example Input:

Case 1: Monthly salary: 100000 (Default tax: 15%)

Case 2: Revenue: 2000000, Expenses: 1200000

Case 3: Earnings: 800000, Dependents: 3

Example Output:

Case 1: Tax = 180000

Case 2: Tax = 120000

Case 3: Tax = 72000

6. Write a program to determine the total number of prime factors of a number using recursion. Create the following functions:
- Recursive function to check if a number is prime:
bool isPrime(int n, int divisor = 2);
 - Recursive function to count prime factors:
int countPrimeFactors(int n, int current = 2);

The program should prompt the user to enter a number and display the total number of its prime factors.

Input:

Number = 56

Output:

Prime factors: 2, 2, 2, 7

Total prime factors = 4